Examining Age-Related Differences in Neural Responses to Trust

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1. Introduction

• Social relationships influence our behaviors and modulate our neural responses to reward (Fareri et al., 2015).

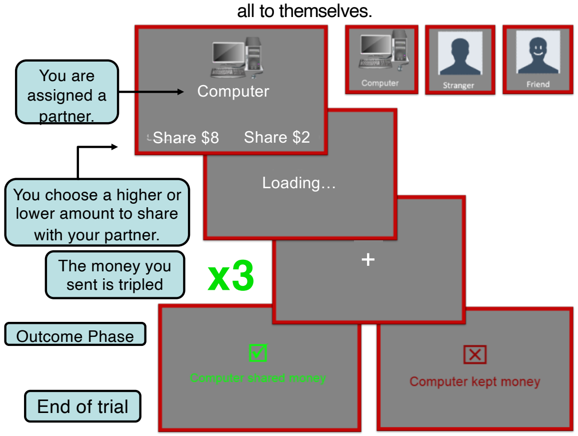
• The effect of relationships on behavior and reward activation change as we age (Fareri et al., 2022).

• It remains unclear whether age-related differences in responses to trust are associated with maladaptive outcomes such as financial exploitation.

We investigated whether social trust decisions change as we age, and if those changes could be related to financial exploitation.

2. Methods

Participants (N = 62) completed a trust task with three partners: a computer, a stranger, and their chosen friend. They are given $8 each round and must share one of two pre-determined amounts. That amount is then tripled, and they are shown whether their partner reciprocated half of the total, or defected and kept it all to themselves.



We used the Inclusion of Other in Self Scale (IOS) and the Older Adults Financial Exploitation Measure (OAFEM) to measure closeness ratings and risk for financial exploitation.

Whole brain results were thresholded at Z > 3.1 and Family Wise Error corrected at p > .05

fMRI data were preprocessed with fMRIprep and statistical analyses were performed using FSL

Scan the QR code to see the Pre-registration!

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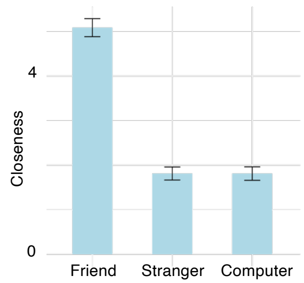
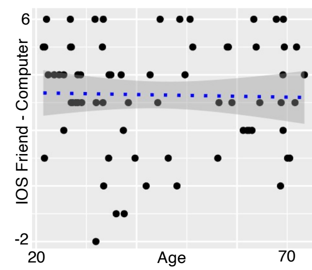
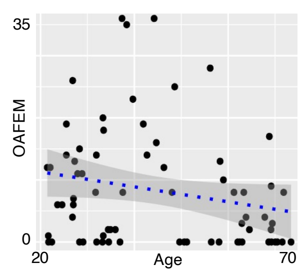
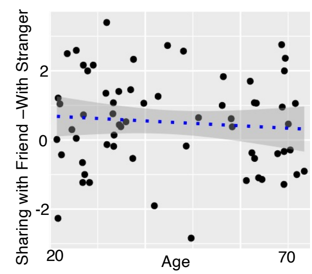
3. Age-Related Behavioral Effects





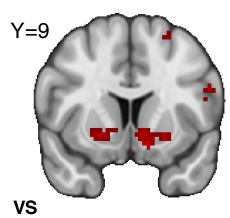
Participants are closer with their friend than with the stranger and computer partners, however, there is no relationship between age and closeness ratings.

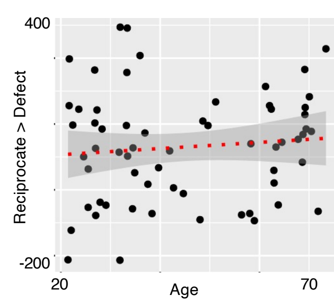




We found no age-related effects across financial exploitation measures or investment behavior

4. Age-Related Differences in Striatal Response?

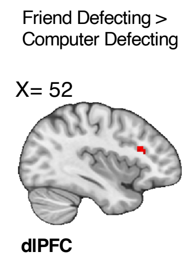


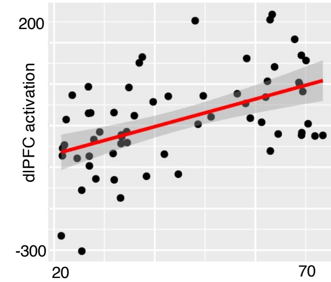


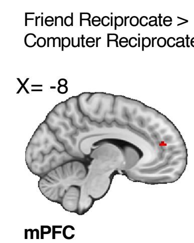
We found activation in the ventral striatum (VS) when participants were presented with a reciprocate outcome as opposed to a defect outcome, however, there was no significant association with age for any condition (Friend > Stranger and Friend > Computer).

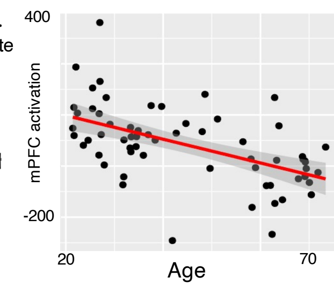
5. Whole Brain Analyses of Neural Responses to Social Outcomes

To investigate if age-related effects were present elsewhere, we conducted a set of whole brain analyses.









We found that friend defecting resulted in increased dorsolateral prefrontal cortex (**dlPFC)** activation in older adults compared to younger adults.

We also found that friend reciprocating resulted in reduced medial prefrontal cortex (**mPFC**) activation in older adults compared to younger adults.

6. Conclusions and Future Directions

• Our preliminary results indicate that **reciprocation elicited activation in the ventral striatum**.

• Additionally, we find that activation of the **dlPFC increased**, and **mPFC decreased,** during the processing of social outcomes **as a function of age**.

• Future analyses will examine how neural and behavioral responses to trust are associated with risk for financial exploitation.